Supplementary Material

LOCUS: Verifying OCR by Local Optimality over Counterfactual String Graphs

Anonymous Author(s)

This document supplements the article *LOCUS: Verifying OCR by Local Optimality over Counterfactual String Graphs*. It gives the implementation details needed for reproduction and the complete experimental record behind the summaries of the main article; references without the prefix S (e.g. "Table 3") point to the main article, and numbered citations such as [6] refer to its reference list. All tables and figures are generated by scripts from the raw result files; no number was entered by hand.

S1. Benchmark construction

Crops. FUNSD words are cropped from the page with a 2-pixel margin after rescaling the LayoutLM-normalised boxes ($[0, 1000]^2$) of the public Hugging Face release to pixel coordinates; cropping with the raw values yields misaligned crops for almost every word (the Tesseract word error rate is 99.8% without the rescaling and 36.9% with it). CORD words are cropped from the axis-aligned bounding box of each word quadrilateral; the official train and validation receipts form the training pool. SROIE, IAM and IIIT5K are distributed as crops. Crops narrower than 6 or shorter than 8 pixels, and empty transcriptions, are discarded.

Engines. Tesseract 5.2 (LSTM engine, `-psm 7`, crops shorter than 32 px are bicubically upscaled to 32 px) outputs the space-joined words it detects and their mean confidence. TrOCR-small (`microsoft/trocr-small-printed` or `-handwritten`) decodes greedily; its confidence is the mean token probability.

Labels. A claim is CORRECT iff it equals the ground truth after whitespace normalisation (case-insensitive for IIIT5K). No other normalisation is applied, so case and punctuation errors count as errors, as they do for any downstream consumer that copies the field verbatim.

Caveats. TrOCR's printed checkpoints were fine-tuned on SROIE receipts; the verifier V_1 is therefore in-domain on SROIE, and TrOCR-small makes only 3–6% errors there, which is why SROIE uses Tesseract claims. Tesseract is degenerate on handwriting (93.9% word error on IAM), which is why IAM uses TrOCR-small claims.

S2. Counterfactual neighbourhood generation

Confusion groups. Substitutions are drawn from the groups `OOoQDCc`, `Il1|!ij`, `5Ss$`, `8B&3`, `2Zz`, `6Gb`, `9gq`, `7T/`, `uvnUV`, `wW`, `.,:;’‘"`, `-_~=`, `(){}[]`, `aeo`, `hbk`, `rnm`, `cC(e`, `xX%`, `yY`, `pP`, `fft`; at most two alternatives per character are used.

Algorithm. Given claim h and verifier readings g_1 (TrOCR) and g_2 (PARSeq):

1. add g_2 and g_1 (if different from h);
2. for each reading, add every single edit of h suggested by its Levenshtein alignment to the reading: substitution of an aligned mismatching character, deletion of an unaligned character of h, insertion of an unaligned character of the reading;
3. for each position of h (up to 32): a case flip, up to two confusion substitutions, and a deletion;
4. deduplicate, keep insertion order, and stop at $M = 28$ neighbours.

No step uses the ground truth. The share of real errors whose truth is a node of the graph is reported in Table 2.

Rendering. $R(v)$ draws v in Liberation Sans (metrically equivalent to Arial) at $0.75\times$ the crop height (clamped to $[32, 96]$ px), black on white, with a quarter-height margin. The same rendering is fed to both verifiers. PARSeq scores the string restricted to its 94-character charset and truncated to its maximum label length of 25.

S3. Label-free pretraining data

From the unlabeled training pool we keep images on which two independent engines produce the identical string (Tesseract and TrOCR-small; TrOCR-small and TrOCR-base on IAM), shuffle them with a fixed seed and keep at most 3,000. Each yields a pseudo-correct graph (the agreed string) and, if the corruption changes the string, a pseudo-incorrect graph built around one or two random OCR-style edits: confusable substitution (45%), case flip (15%), deletion (20%) or insertion of a confusable character (20%). The repair target of a corrupted graph is the node holding the agreed string (when present). Agreement is a proxy, not ground truth; its precision on the training pool is reported in Table S1.

Table S1. Label-free pretraining sets. Precision of engine agreement is measured post hoc against the ground truth of the training pool (never used for training).

Dataset	agreeing images	graphs	precision of agreement	truth in graph (corruptions)
FUNSD	2,561	5,085	94.5%	99.6%
CORD	3,000	5,959	98.7%	99.8%
SROIE	2,663	5,301	99.0%	100.0%
IAM	3,000	5,950	83.7%	100.0%
IIIT5K	860	1,711	99.3%	99.5%

S4. Hyperparameters and baseline protocols

Table S2. Hyperparameters (identical for all datasets and all k).

Node features	2×11 evidence + 7 claim deltas (tanh-squashed) + 8 edit-kind + 1 position + 6 agreement (claim node)
Edge features	8 edit kinds + sibling + path + self (one-hot)
GNN	3 GATv2 layers, hidden 64, 4 heads, residual + LayerNorm, dropout 0.2
Readout	claim state, attention pool, max pool over rivals, $\beta \log P_{\text{opt}}$ (β learned)
Pretraining	15 epochs, AdamW lr 10^{-3}, wd 10^{-4}, batch 64, $\lambda = 0.5$
Few-shot fine-tuning	300 steps, AdamW lr $3 \cdot 10^{-4}$, batch $\min(64, k)$, class-balanced BCE
Head selection	repeated stratified CV (3 repeats, $\leq$5 folds) on the support only; fusion if min class < 5
Anchored head	priors scaled $\times 10$; residual ℓ_2 LR, $C \in \{.003, .01, .03, .1, .3\}$ by CV on the support
Full supervision	60 epochs, early stopping (patience 8) on a 15% split of the training pool
Seeds	0–4 (support sampling, initialisation, pretraining)

Thresholds. Engine confidence and the ρ-margin rule are thresholded at the value maximising macro-F1 on the k-shot support (seed-specific). Agreement baselines are binary (accept iff equal). The graded PARSeq score is $+c$ if PARSeq's reading equals the claim and $-c$ otherwise, where c is PARSeq's sequence confidence (product of character probabilities); it is thresholded at 0, which coincides with the binary decision.

VLM judge prompt. *Does the image show exactly the text "{claim}" (same characters, case and punctuation)? Answer yes or no.* The image is passed at its native resolution (upscaled to a height of 56 px if smaller; Qwen2.5-VL pixel budget 32–256 visual tokens). The score is the renormalised probability of `yes`/`Yes` versus `no`/`No` as the first generated token.

ConfidNet-style learned confidence. A one-hidden-layer MLP (32 units, ℓ_2 10^{-2}) on the engine confidence and five likelihood statistics of the claim under TrOCR-base, trained on the support set.

S5. Additional analyses and qualitative examples

Error types

Table S3 breaks down the detection rate (recall of truly incorrect claims) by the edit that separates the claim from the truth. Operating points: LOCUS at $P = 0.5$, PARSeq-graded at its natural threshold, the VLM at 0.5, and engine confidence at its *test-optimal* threshold (favourable to the baseline).

Table S3. Detection rate (%) by error type; number of errors of that type in parentheses ($k = 25$).

Dataset	Method	substitution	insertion	deletion	case only	multi-edit
FUNSD	LOCUS	87.7 (236)	79.7 (390)	71.8 (78)	77.2 (57)	97.1 (713)
	PARSeq re-OCR (graded)	86.0 (236)	80.3 (390)	70.5 (78)	78.9 (57)	98.2 (713)
	TrOCR-base re-OCR	93.2 (236)	85.1 (390)	82.1 (78)	96.5 (57)	98.5 (713)
	Qwen2.5-VL-7B judge	62.3 (236)	37.7 (390)	67.9 (78)	26.3 (57)	79.2 (713)
	Engine confidence	74.2 (236)	76.4 (390)	48.7 (78)	71.9 (57)	89.3 (713)
CORD	LOCUS	95.3 (342)	93.2 (73)	92.5 (40)	80.0 (20)	99.1 (466)
	PARSeq re-OCR (graded)	95.6 (342)	98.6 (73)	95.0 (40)	100.0 (20)	100.0 (466)
	TrOCR-base re-OCR	98.0 (342)	100.0 (73)	95.0 (40)	90.0 (20)	100.0 (466)
	Qwen2.5-VL-7B judge	74.3 (342)	20.5 (73)	87.5 (40)	10.0 (20)	94.2 (466)
	Engine confidence	64.6 (342)	53.4 (73)	92.5 (40)	55.0 (20)	92.9 (466)
SROIE	LOCUS	97.5 (641)	76.6 (137)	72.4 (105)	–	99.3 (997)
	PARSeq re-OCR (graded)	98.9 (641)	100.0 (137)	79.0 (105)	–	99.8 (997)
	TrOCR-base re-OCR	99.5 (641)	82.5 (137)	79.0 (105)	–	99.4 (997)
	Qwen2.5-VL-7B judge	68.2 (641)	29.2 (137)	37.1 (105)	–	90.4 (997)
	Engine confidence	75.5 (641)	75.9 (137)	44.8 (105)	–	92.8 (997)
IAM	LOCUS	93.5 (505)	76.8 (95)	57.7 (26)	43.2 (185)	92.8 (1046)
	PARSeq re-OCR (graded)	95.6 (505)	98.9 (95)	57.7 (26)	94.6 (185)	98.1 (1046)
	TrOCR-base re-OCR	88.9 (505)	85.3 (95)	76.9 (26)	29.2 (185)	95.9 (1046)
	Qwen2.5-VL-7B judge	80.0 (505)	61.1 (95)	61.5 (26)	57.8 (185)	86.3 (1046)
	Engine confidence	82.8 (505)	68.4 (95)	69.2 (26)	27.0 (185)	89.5 (1046)
IIIT5K	LOCUS	92.4 (172)	96.4 (28)	95.1 (61)	–	99.8 (608)
	PARSeq re-OCR (graded)	94.2 (172)	100.0 (28)	95.1 (61)	–	99.8 (608)
	TrOCR-base re-OCR	93.6 (172)	100.0 (28)	93.4 (61)	–	99.8 (608)
	Qwen2.5-VL-7B judge	89.0 (172)	96.4 (28)	93.4 (61)	–	99.7 (608)
	Engine confidence	50.6 (172)	60.7 (28)	65.6 (61)	–	94.1 (608)



Figure S1. AUROC by claim length ($k = 25$). Bins with fewer than 30 words or a single class are omitted.

Table S4. Expected calibration error (15 bins) of the correctness probability ($k = 25$; PARSeq-graded mapped to $[0,1]$ as $(s+1)/2$). Lower is better.

Dataset	LOCUS	PARSeq graded	Qwen2.5-VL-7B	Engine conf.
FUNSD	0.024	0.039	0.019	0.136
CORD	0.025	0.018	0.045	0.114
SROIE	0.020	0.197	0.057	0.213
IAM	0.030	0.058	0.110	0.289
IIIT5K	0.027	0.009	0.136	0.042

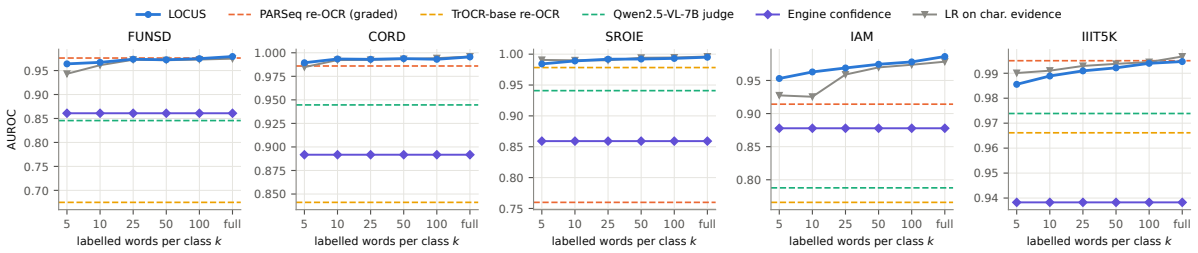


Figure S2. AUROC as a function of the number of labelled words per class (ensembles of five seeds; dashed: label-free baselines).

Table S5. Cross-dataset transfer, AUROC (rows: training dataset, fully supervised; columns: test dataset; diagonal: in-domain).

Train \ Test	FUNSD	CORD	SROIE	IAM	IIIT5K
FUNSD	.979	.995	.979	.968	.993
CORD	.968	.996	.987	.954	.986
SROIE	.941	.992	.996	.892	.992
IAM	.962	.989	.952	.985	.991
IIIT5K	.961	.993	.975	.963	.995

Table S6. Ablations at $k = 25$, AUROC (ensemble of five seeds). Last column: mean change w.r.t. the full model.

Variant	FUNSD	CORD	SROIE	IAM	IIIT5K	Δ
LOCUS (full)	.973	.993	**.992**	.968	.991	
+ CV-selected prior/anchored head	.974	.993	.990	.968	.992	-0.000
+ linear probe on embedding	.974	.993	**.992**	.969	**.993**	+0.001
w/o verifier-agreement node features	.974	.993	.991	.968	.987	-0.001
w/o edit-graph GNN (char evidence LR)	.973	.992	.990	.959	**.993**	-0.002
w/o label-free pretraining	.969	.992	.990	**.970**	.990	-0.002
w/o local-optimality term	.972	.993	**.992**	.966	.990	-0.001
w/o repair loss	.971	.993	**.992**	.965	.990	-0.001
w/o render-normalisation	.973	.993	**.992**	.967	.992	-0.000
w/o latent render distance	.973	.993	**.992**	.969	.990	-0.000
likelihood only (no render)	.974	.992	.991	.969	.992	+0.000
TrOCR verifier only	.964	.991	**.992**	.956	.991	-0.005
PARSeq verifier only	.974	.993	.990	.965	.992	-0.001
star edges only	.971	.993	**.992**	.969	.991	-0.001
no message passing	**.975**	**.994**	**.992**	.967	.991	+0.000
GCN backbone	.970	.993	.991	.966	.990	-0.002
GraphSAGE backbone	.973	.993	**.992**	**.970**	.990	-0.000
GIN backbone	.973	.991	.989	.957	.990	-0.004

Claim length

Calibration

AUROC and AURC learning curves

Cross-dataset transfer (AUROC)

Ablations: AUROC and repair

Sensitivity

Table S7. Sensitivity of LOCUS ($k = 25$, ensemble of five seeds) to the size of the label-free pretraining set, the number of fine-tuning steps, and the support-selection strategy (label-free k-means clusters the unlabeled pool in the pretrained embedding space and labels the $2k$ medoids).

		FUNSD		CORD		IAM	
Factor	Value	F1	AUROC	F1	AUROC	F1	AUROC
pretrain fraction	0.1	.943	.967	.977	.993	.902	.968
pretrain fraction	0.25	**.944**	.968	.978	**.994**	**.908**	**.970**
pretrain fraction	0.5	.941	.967	.979	**.994**	.904	**.970**
pretrain fraction	1.0	.940	**.974**	**.980**	.993	.900	.967
fine-tuning steps	100	**.942**	**.976**	.975	**.993**	.884	.960
fine-tuning steps	300	.940	.974	**.980**	**.993**	**.900**	**.967**
fine-tuning steps	1000	.938	.974	.979	**.993**	.899	**.967**
support selection	random	**.940**	**.974**	**.980**	.993	.900	**.967**
support selection	label-free k-means	.935	.964	**.980**	**.994**	**.907**	.964

Main comparison at other label budgets and metrics

Composition of real OCR errors

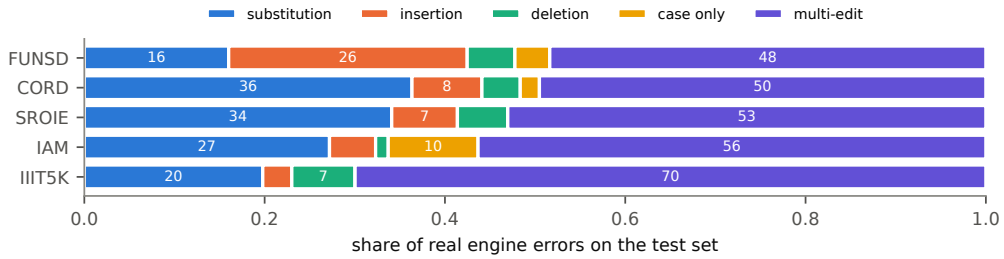


Figure S3. Types of real engine errors on each test set (share of errors; labels for shares above 7%). Handwriting and receipt lines are dominated by multi-edit errors, which bounds what a single-edit neighbourhood can repair (Table 2).

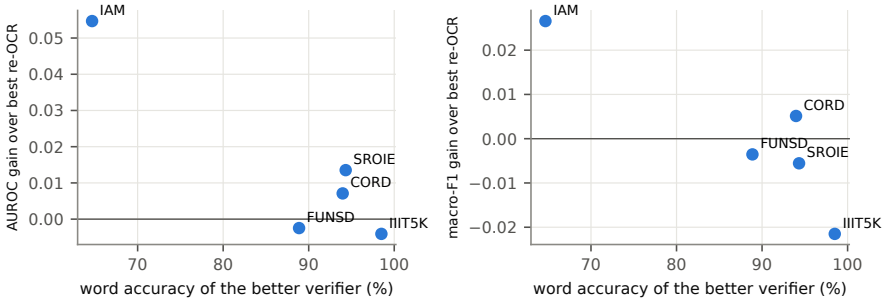


Figure S4. Gain of LOCUS over the best re-recognition baseline against the word accuracy of the better verifier (one point per dataset).

S6. Full results

Tables S16–S30 report every method at every k with seed standard deviations, and every metric at $k = 25$.

Table S8. Macro-F1 at $k = 10$ (protocol of Table 3).

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.722	.775	.749	.782	.817	.769
ConfidNet-style learned conf. [9]	.702	.813	.785	.783	.915	.800
Two-engine voting	.609	.772	.976	.759	.862	.796
Re-OCR: TrOCR-base [5]	.605	.811	**.979**	.758	.952	.821
Re-OCR: Florence-2 [6]	.647	.615	.478	.369	.823	.586
Re-OCR: PARSeq [4]	**.946**	.973	.742	.879	**.993**	.907
Re-OCR: PARSeq, graded	**.946**	.973	.742	.879	**.993**	.907
VLM judge: Qwen2.5-VL-7B [7]	.760	.891	.866	.700	.888	.821
VLM judge, oracle threshold†	.762	.902	.883	.719	.928	.839
Best baseline per dataset, chosen in hindsight‡	.946	.973	.979	.879	.993	.954
Strong learned baselines on our evidence						
LR on flattened graph evidence	.931	.957	.939	.881	.974	.937
LR on character evidence	.915	.970	.949	.839	.965	.927
GBDT on all evidence	.885	.960	.967	.882	.947	.928
DeepSets over rivals (no edges)	.931	.978	.930	**.896**	.964	.940
LOCUS-GNN from scratch	.920	.969	.913	.877	.961	.928
Ours						
LOCUS zero-shot (no labels)	.918	.904	.798	.825	.968	.883
LOCUS (k-shot)	.936	**.980**	.954	.888	.975	**.947**

Table S9. Macro-F1 at $k = 100$.

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.771	.799	.777	.801	.790	.788
ConfidNet-style learned conf. [9]	.830	.905	.873	.832	.924	.873
Two-engine voting	.609	.772	.976	.759	.862	.796
Re-OCR: TrOCR-base [5]	.605	.811	**.979**	.758	.952	.821
Re-OCR: Florence-2 [6]	.647	.615	.478	.369	.823	.586
Re-OCR: PARSeq [4]	**.946**	.973	.742	.879	**.993**	.907
Re-OCR: PARSeq, graded	**.946**	.973	.742	.879	**.993**	.907
VLM judge: Qwen2.5-VL-7B [7]	.760	.891	.866	.700	.888	.821
VLM judge, oracle threshold†	.762	.902	.883	.719	.928	.839
Best baseline per dataset, chosen in hindsight‡	.946	.973	.979	.879	.993	.954
Strong learned baselines on our evidence						
LR on flattened graph evidence	.945	**.980**	.978	.918	.970	.958
LR on character evidence	.940	.979	.977	.920	.967	.956
GBDT on all evidence	.944	**.980**	**.979**	.923	.983	**.962**
DeepSets over rivals (no edges)	.942	**.980**	.976	.926	.968	.958
LOCUS-GNN from scratch	.938	.978	.977	**.927**	.973	.959
Ours						
LOCUS zero-shot (no labels)	.918	.904	.798	.825	.968	.883
LOCUS (k-shot)	**.946**	**.980**	.974	**.927**	.973	.960

Table S10. Macro-F1 with full supervision (85% of the training pool; 15% for early stopping).

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.787	.805	.790	.807	.854	.809
ConfidNet-style learned conf. [9]	.856	.934	.912	.859	.957	.904
Two-engine voting	.609	.772	.976	.759	.862	.796
Re-OCR: TrOCR-base [5]	.605	.811	**.979**	.758	.952	.821
Re-OCR: Florence-2 [6]	.647	.615	.478	.369	.823	.586
Re-OCR: PARSeq [4]	.946	.973	.742	.879	**.993**	.907
Re-OCR: PARSeq, graded	.946	.973	.742	.879	**.993**	.907
VLM judge: Qwen2.5-VL-7B [7]	.760	.891	.866	.700	.888	.821
VLM judge, oracle threshold†	.762	.902	.883	.719	.928	.839
Best baseline per dataset, chosen in hindsight‡	.946	.973	.979	.879	.993	.954
Strong learned baselines on our evidence						
LR on flattened graph evidence	.949	.977	.978	.934	.976	.963
LR on character evidence	.948	.980	**.979**	.928	.976	.962
GBDT on all evidence	**.951**	.980	.978	.944	.989	.968
DeepSets over rivals (no edges)	.949	**.983**	.978	.947	.982	.968
LOCUS-GNN from scratch	**.951**	**.983**	**.979**	**.950**	.987	**.970**
Ours						
LOCUS zero-shot (no labels)	.918	.904	.798	.825	.968	.883
LOCUS (k-shot)	.950	.981	**.979**	.945	.985	.968

Table S11. AUROC at $k = 10$.

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.861	.892	.859	.878	.938	.886
ConfidNet-style learned conf. [9]	.763	.907	.867	.849	.981	.873
Two-engine voting	.680	.807	.976	.767	.905	.827
Re-OCR: TrOCR-base [5]	.675	.841	.978	.766	.966	.845
Re-OCR: Florence-2 [6]	.715	.685	.578	.522	.879	.676
Re-OCR: PARSeq [4]	.939	.975	.763	.885	.991	.911
Re-OCR: PARSeq, graded	**.976**	.986	.760	.914	**.995**	.926
VLM judge: Qwen2.5-VL-7B [7]	.846	.945	.941	.788	.974	.899
VLM judge, oracle threshold†	.846	.945	.941	.788	.974	.899
Best baseline per dataset, chosen in hindsight‡	.976	.986	.978	.914	.995	.970
Strong learned baselines on our evidence						
LR on flattened graph evidence	.972	**.995**	**.991**	.961	.991	**.982**
LR on character evidence	.961	.992	.990	.925	.991	.972
GBDT on all evidence	.952	.988	.986	.948	.982	.971
DeepSets over rivals (no edges)	.960	**.995**	.985	.960	.988	.978
LOCUS-GNN from scratch	.955	.992	.977	.957	.987	.974
Ours						
LOCUS zero-shot (no labels)	.949	.977	.966	.909	.983	.957
LOCUS (k-shot)	.967	.993	.989	**.963**	.989	.980

Table S12. AUROC at $k = 100$.

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.861	.892	.859	.878	.938	.886
ConfidNet-style learned conf. [9]	.907	.964	.948	.894	.985	.940
Two-engine voting	.680	.807	.976	.767	.905	.827
Re-OCR: TrOCR-base [5]	.675	.841	.978	.766	.966	.845
Re-OCR: Florence-2 [6]	.715	.685	.578	.522	.879	.676
Re-OCR: PARSeq [4]	.939	.975	.763	.885	.991	.911
Re-OCR: PARSeq, graded	**.976**	.986	.760	.914	.995	.926
VLM judge: Qwen2.5-VL-7B [7]	.846	.945	.941	.788	.974	.899
VLM judge, oracle threshold†	.846	.945	.941	.788	.974	.899
Best baseline per dataset, chosen in hindsight‡	.976	.986	.978	.914	.995	.970
Strong learned baselines on our evidence						
LR on flattened graph evidence	.975	**.996**	.994	.976	.993	**.987**
LR on character evidence	.973	.994	**.995**	.974	.994	.986
GBDT on all evidence	.975	.995	**.995**	.976	**.996**	**.987**
DeepSets over rivals (no edges)	.972	.995	.993	**.979**	.995	**.987**
LOCUS-GNN from scratch	.970	.994	.993	.977	.995	.986
Ours						
LOCUS zero-shot (no labels)	.949	.977	.966	.909	.983	.957
LOCUS (k-shot)	.975	.993	.993	.978	.994	**.987**

Table S13. AURC at $k = 25$ (lower is better; best = lowest).

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.144	.140	.211	.203	.069	.153
ConfidNet-style learned conf. [9]	.164	.124	.184	.226	.054	.150
Two-engine voting	.220	.170	.152	.272	.076	.178
Re-OCR: TrOCR-base [5]	.229	.154	.151	.273	.058	.173
Re-OCR: Florence-2 [6]	.198	.233	.377	.424	.083	.263
Re-OCR: PARSeq [4]	.125	.109	.246	.186	.050	.143
Re-OCR: PARSeq, graded	**.090**	.099	.242	.168	**.049**	.130
VLM judge: Qwen2.5-VL-7B [7]	.153	.127	.172	.250	.056	.151
VLM judge, oracle threshold†	.153	.127	.172	.250	.056	.151
Best baseline per dataset, chosen in hindsight‡	.090	.099	.151	.168	.049	.111
Strong learned baselines on our evidence						
LR on flattened graph evidence	.092	**.096**	.139	**.146**	.052	**.105**
LR on character evidence	.094	.098	.139	.152	.050	.107
GBDT on all evidence	.098	.100	**.138**	.150	.052	.108
DeepSets over rivals (no edges)	.094	.097	**.138**	.147	.052	.106
LOCUS-GNN from scratch	.095	.098	.139	.149	.052	.107
Ours						
LOCUS zero-shot (no labels)	.110	.104	.152	.180	.054	.120
LOCUS (k-shot)	.092	.097	.139	.149	.052	.106

Table S14. Coverage at 95% accepted precision at $k = 25$ (higher is better).

Method	FUNSD	CORD	SROIE	IAM	IIIT5K	Avg.
Baselines						
Engine MSP confidence	.346	.408	.091	.091	.631	.313
ConfidNet-style learned conf. [9]	.053	.465	.233	.007	.726	.297
Two-engine voting	.021	.409	.547	.001	.635	.322
Re-OCR: TrOCR-base [5]	.000	.447	.550	.001	.715	.343
Re-OCR: Florence-2 [6]	.311	.249	.092	.001	.595	.249
Re-OCR: PARSeq [4]	.006	.615	.305	.455	.746	.426
Re-OCR: PARSeq, graded	**.646**	.614	.304	.454	**.747**	.553
VLM judge: Qwen2.5-VL-7B [7]	.147	.351	.238	.052	.716	.301
VLM judge, oracle threshold†	.147	.351	.238	.052	.716	.301
Best baseline per dataset, chosen in hindsight‡	.646	.615	.550	.455	.747	.603
Strong learned baselines on our evidence						
LR on flattened graph evidence	.645	.625	.546	.483	.745	.609
LR on character evidence	.631	.624	.545	.446	.745	.598
GBDT on all evidence	.625	.621	.548	.467	.746	.601
DeepSets over rivals (no edges)	.639	**.627**	.552	**.497**	**.747**	**.612**
LOCUS-GNN from scratch	.630	.624	.550	.482	**.747**	.607
Ours						
LOCUS zero-shot (no labels)	.270	.551	.401	.346	.739	.461
LOCUS (k-shot)	.628	.626	**.553**	.486	.744	.607

Table S15. Single-verifier setting (no agreement features), $k = 25$, macro-F1: effect of render normalisation when only one recogniser is available.

Variant	FUNSD	CORD	SROIE	IAM	IIIT5K
TrOCR only	.930	**.973**	**.956**	.839	**.957**
TrOCR only, w/o render-norm.	.929	.967	.902	.849	.951
PARSeq only	.932	.968	.918	**.885**	.943
PARSeq only, w/o render-norm.	**.933**	.969	.916	.884	.947

Table S16. FUNSD: macro-f1 as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.726	.722	.756	.764	.771	.787
ConfidNet-style	.683	.702	.771	.812	.830	.856
Flat-LR	$.897_{.041}$	$.931_{.032}$	$\mathbf{.948}_{.009}$	$.946_{.007}$	$.945_{.011}$	$.949_{.000}$
CharEv-LR	$.870_{.033}$	$.915_{.039}$	$.935_{.005}$	$.937_{.001}$	$.940_{.002}$	$.948_{.001}$
GBDT	$.874_{.061}$	$.885_{.070}$	$.922_{.034}$	$.943_{.006}$	$.944_{.008}$	$\mathbf{.951}_{.001}$
DeepSets	$.898_{.021}$	$.931_{.063}$	$.942_{.042}$	$\mathbf{.947}_{.009}$	$.942_{.016}$	$.949_{.001}$
LOCUS-GNN scratch	$.868_{.033}$	$.920_{.060}$	$.933_{.031}$	$.944_{.014}$	$.938_{.022}$	$\mathbf{.951}_{.001}$
LOCUS probe	$.921_{.013}$	$.938_{.009}$	$.946_{.014}$	$\mathbf{.947}_{.003}$	$\mathbf{.947}_{.007}$	$.950_{.001}$
LOCUS + CV head	$.929_{.011}$	$.934_{.008}$	$.942_{.023}$	$.946_{.004}$	$.945_{.010}$	$.949_{.001}$
Priors-only LR	$\mathbf{.940}_{.005}$	$\mathbf{.946}_{.002}$	$.947_{.001}$	$\mathbf{.947}_{.001}$	$\mathbf{.947}_{.000}$	$.946_{.000}$
LOCUS	$.923_{.020}$	$.936_{.030}$	$.943_{.007}$	$.943_{.009}$	$.946_{.004}$	$.950_{.002}$

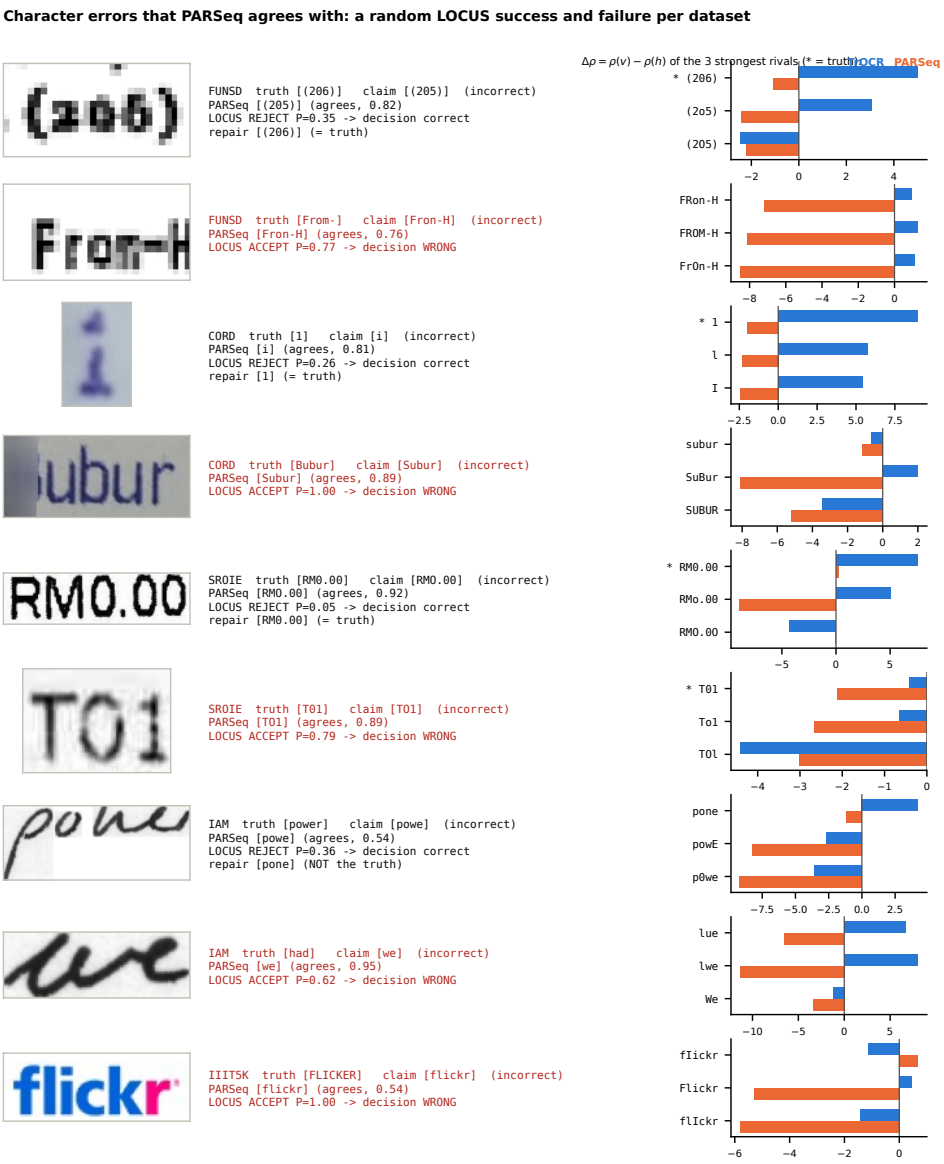


Figure S5. Errors that PARSeq re-recognition accepts (it reads the same string as the engine), restricted to genuine character errors (case-only and punctuation-only differences excluded): for each dataset one random LOCUS success and one random failure. LOCUS rejects only a small fraction of this stratum (Table 15); the manual audit (Table 16) shows why: most of these "errors" are annotation or cropping problems, i.e. the claim matches what is visible.

Table S17. CORD: macro-f1 as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.742	.775	.777	.793	.799	.805
ConfidNet-style	.751	.813	.842	.882	.905	.934
Flat-LR	$.933_{.019}$	$.957_{.018}$	$.955_{.018}$	$.963_{.012}$	$\mathbf{.980}_{.004}$	$.977_{.001}$
CharEv-LR	$.940_{.044}$	$.970_{.016}$	$.971_{.016}$	$.975_{.004}$	$\underline{.979}_{.003}$	$.980_{.000}$
GBDT	$.917_{.064}$	$.960_{.016}$	$.961_{.015}$	$.970_{.008}$	$\mathbf{.980}_{.004}$	$.980_{.001}$
DeepSets	$\underline{.949}_{.033}$	$\underline{.978}_{.019}$	$\underline{.977}_{.015}$	$\underline{.980}_{.002}$	$\mathbf{.980}_{.003}$	$\mathbf{.983}_{.001}$
LOCUS-GNN scratch	$.916_{.013}$	$.969_{.025}$	$.969_{.020}$	$.977_{.002}$	$.978_{.003}$	$\mathbf{.983}_{.001}$
LOCUS probe	$.948_{.024}$	$.953_{.019}$	$.961_{.020}$	$\mathbf{.981}_{.006}$	$\mathbf{.980}_{.002}$	$\underline{.981}_{.001}$
LOCUS + CV head	$.926_{.018}$	$.942_{.025}$	$.962_{.023}$	$.978_{.008}$	$\underline{.979}_{.002}$	$\underline{.981}_{.000}$
Priors-only LR	$\mathbf{.978}_{.002}$	$\underline{.978}_{.000}$	$\mathbf{.978}_{.000}$	$.978_{.000}$	$.978_{.000}$	$.978_{.000}$
LOCUS	$.943_{.019}$	$\mathbf{.980}_{.024}$	$\mathbf{.978}_{.023}$	$\mathbf{.981}_{.002}$	$\mathbf{.980}_{.002}$	$\underline{.981}_{.003}$

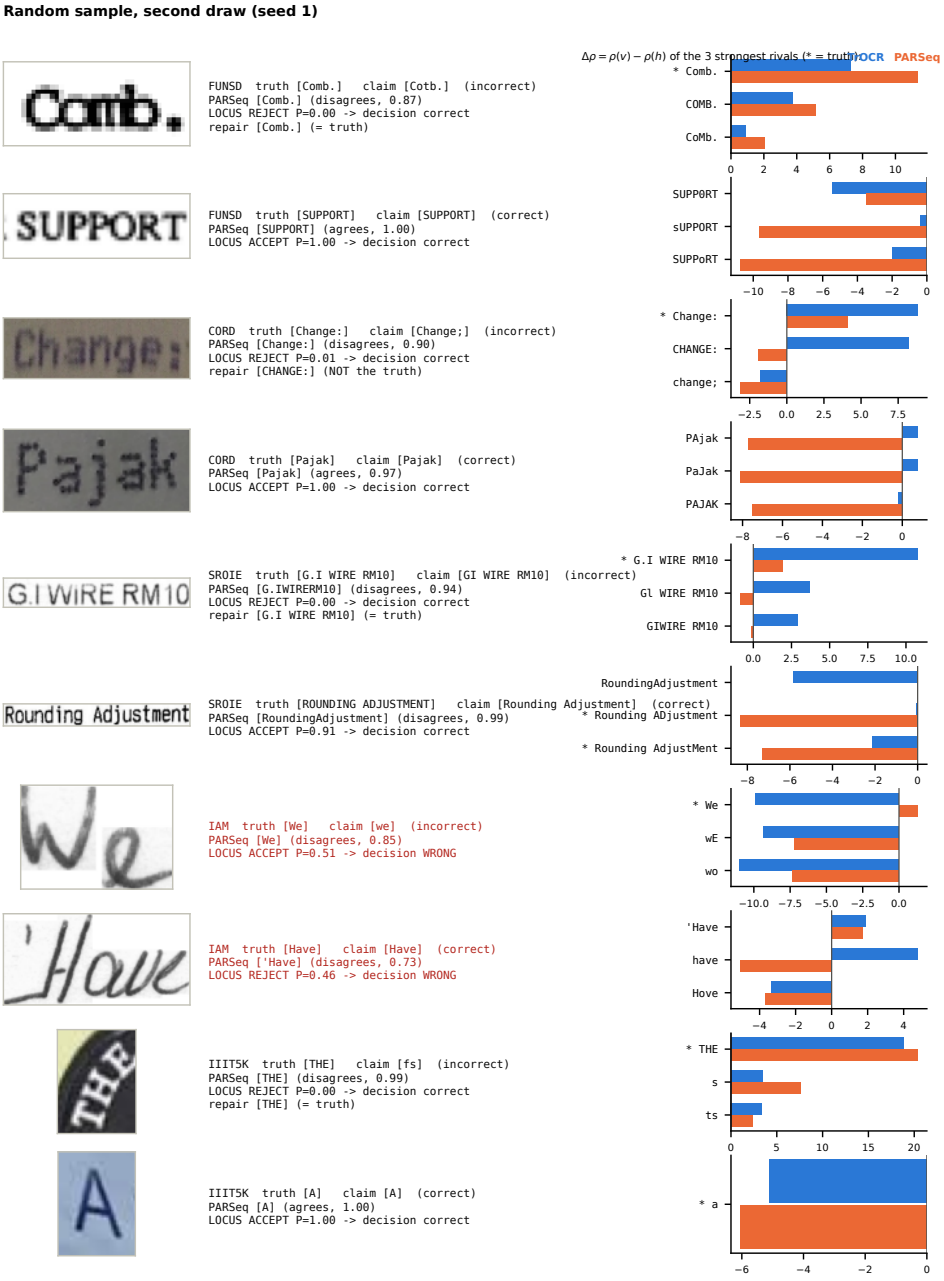


Figure S6. Second random sample (seed 1), same protocol as Fig. 8.

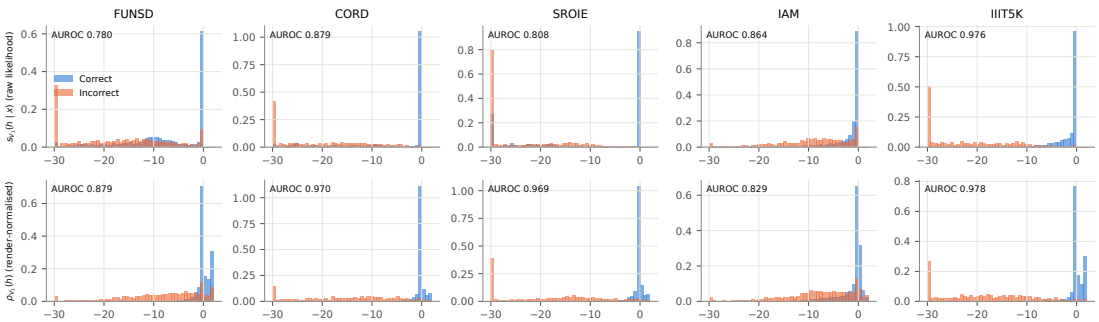


Figure S7. Render normalisation on real data. Distribution of the claim's raw TrOCR likelihood (top) and of its render-normalised ratio ρ (bottom) for correct and incorrect claims, with the AUROC of each single score.

Table S18. SROIE: macro-f1 as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.758	.749	.780	.777	.777	.790
ConfidNet-style	.746	.785	.826	.863	.873	.912
Flat-LR	$.932_{.026}$	$.939_{.011}$	$.955_{.018}$	$.978_{.020}$	$.978_{.007}$	$.978_{.000}$
CharEv-LR	$.941_{.084}$	$.949_{.022}$	$.959_{.013}$	$.973_{.006}$	$.977_{.004}$	$\mathbf{.979}_{.000}$
GBDT	$.956_{.019}$	$.967_{.029}$	$.962_{.035}$	$.978_{.015}$	$\mathbf{.979}_{.003}$	$.978_{.001}$
DeepSets	$.924_{.029}$	$.930_{.025}$	$.968_{.013}$	$.969_{.007}$	$.976_{.005}$	$.978_{.000}$
LOCUS-GNN scratch	$.903_{.042}$	$.913_{.018}$	$.961_{.018}$	$.970_{.008}$	$.977_{.009}$	$\mathbf{.979}_{.001}$
LOCUS probe	$.961_{.028}$	$.957_{.004}$	$.973_{.007}$	$.973_{.006}$	$.975_{.004}$	$\mathbf{.979}_{.000}$
LOCUS + CV head	$.884_{.051}$	$.956_{.020}$	$.964_{.015}$	$.975_{.004}$	$.977_{.004}$	$.978_{.000}$
Priors-only LR	$\mathbf{.978}_{.034}$	$\mathbf{.978}_{.000}$	$\mathbf{.978}_{.004}$	$\mathbf{.979}_{.003}$	$\mathbf{.979}_{.000}$	$\mathbf{.979}_{.000}$
LOCUS	$.941_{.046}$	$.954_{.017}$	$.973_{.007}$	$.973_{.003}$	$.974_{.003}$	$\mathbf{.979}_{.001}$

Table S19. IAM: macro-f1 as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.700	.782	.802	.801	.801	.807
ConfidNet-style	.762	.783	.801	.812	.832	.859
Flat-LR	$.863_{.038}$	$.881_{.012}$	$.904_{.017}$	$.915_{.009}$	$.918_{.003}$	$.934_{.001}$
CharEv-LR	$.838_{.020}$	$.839_{.015}$	$.887_{.016}$	$.902_{.007}$	$.920_{.008}$	$.928_{.001}$
GBDT	$.840_{.025}$	$.882_{.053}$	$.893_{.030}$	$.920_{.013}$	$.923_{.019}$	$.944_{.002}$
DeepSets	$.873_{.054}$	$.896_{.031}$	$\mathbf{.909}_{.019}$	$.921_{.009}$	$.926_{.009}$	$.947_{.004}$
LOCUS-GNN scratch	$.865_{.039}$	$.877_{.019}$	$.908_{.023}$	$.917_{.012}$	$.927_{.008}$	$\mathbf{.950}_{.004}$
LOCUS probe	$.870_{.037}$	$.875_{.025}$	$.904_{.016}$	$.922_{.008}$	$\mathbf{.931}_{.007}$	$.944_{.003}$
LOCUS + CV head	$.851_{.026}$	$.888_{.030}$	$.905_{.009}$	$.916_{.009}$	$.925_{.009}$	$.944_{.003}$
Priors-only LR	$\mathbf{.874}_{.017}$	$\mathbf{.897}_{.013}$	$.898_{.013}$	$.908_{.003}$	$.907_{.002}$	$.907_{.001}$
LOCUS	$.869_{.020}$	$.888_{.023}$	$.906_{.011}$	$\mathbf{.923}_{.010}$	$.927_{.006}$	$.945_{.003}$

Table S20. IIIT5K: macro-f1 as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.806	.817	.788	.817	.790	.854
ConfidNet-style	.925	.915	.922	.929	.924	.957
Flat-LR	$.966_{.021}$	$.974_{.008}$	$.969_{.005}$	$.967_{.007}$	$.970_{.009}$	$.976_{.003}$
CharEv-LR	$.958_{.022}$	$.965_{.006}$	$.965_{.008}$	$.970_{.004}$	$.967_{.008}$	$.976_{.003}$
GBDT	$.941_{.007}$	$.947_{.032}$	$.955_{.019}$	$.968_{.039}$	$\mathbf{.983}_{.014}$	$\mathbf{.989}_{.002}$
DeepSets	$.964_{.025}$	$.964_{.030}$	$.965_{.015}$	$.968_{.009}$	$.968_{.013}$	$.982_{.008}$
LOCUS-GNN scratch	$.961_{.020}$	$.961_{.046}$	$.962_{.022}$	$.969_{.009}$	$.973_{.009}$	$.987_{.002}$
LOCUS probe	$.966_{.010}$	$.967_{.005}$	$.968_{.008}$	$.970_{.008}$	$.966_{.011}$	$.978_{.004}$
LOCUS + CV head	$.963_{.010}$	$.966_{.008}$	$.971_{.005}$	$.969_{.011}$	$.973_{.012}$	$.978_{.003}$
Priors-only LR	$\mathbf{.983}_{.005}$	$\mathbf{.982}_{.004}$	$\mathbf{.981}_{.002}$	$\mathbf{.980}_{.003}$	$.980_{.003}$	$.977_{.003}$
LOCUS	$.974_{.003}$	$.975_{.030}$	$.971_{.017}$	$.972_{.005}$	$.973_{.010}$	$.985_{.003}$

Table S21. FUNSD: auroc as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.861	.861	.861	.861	.861	.861
ConfidNet-style	.763	.763	.845	.883	.907	.935
Flat-LR	$\mathbf{.965}_{.013}$	$\mathbf{.972}_{.024}$	$\mathbf{.975}_{.003}$	$.974_{.005}$	$.975_{.006}$	$.977_{.000}$
CharEv-LR	$.943_{.034}$	$.961_{.035}$	$.973_{.003}$	$.972_{.004}$	$.973_{.003}$	$.975_{.000}$
GBDT	$.934_{.060}$	$.952_{.030}$	$.969_{.014}$	$.974_{.005}$	$.975_{.006}$	$.979_{.001}$
DeepSets	$.959_{.031}$	$.960_{.056}$	$.971_{.021}$	$.971_{.004}$	$.972_{.004}$	$.979_{.002}$
LOCUS-GNN scratch	$.954_{.020}$	$.955_{.047}$	$.969_{.010}$	$.968_{.008}$	$.970_{.005}$	$.979_{.001}$
LOCUS probe	$.956_{.014}$	$.969_{.013}$	$\mathbf{.975}_{.003}$	$\mathbf{.976}_{.005}$	$\mathbf{.976}_{.006}$	$\mathbf{.980}_{.001}$
LOCUS + CV head	$.948_{.005}$	$.966_{.005}$	$.972_{.004}$	$.975_{.007}$	$.975_{.008}$	$.979_{.001}$
Priors-only LR	$.958_{.002}$	$.967_{.002}$	$.966_{.004}$	$.966_{.004}$	$.967_{.002}$	$.968_{.001}$
LOCUS	$.964_{.010}$	$.967_{.011}$	$.974_{.004}$	$.972_{.007}$	$.975_{.003}$	$.979_{.001}$

Table S22. CORD: auroc as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.892	.892	.892	.892	.892	.892
ConfidNet-style	.827	.907	.929	.953	.964	.983
Flat-LR	**.994**$_{.014}$	**.995**$_{.002}$	**.995**$_{.002}$	**.995**$_{.000}$	**.996**$_{.001}$	**.997**$_{.000}$
CharEv-LR	.985$_{.015}$	.992$_{.004}$	.992$_{.001}$	.993$_{.001}$	.994$_{.001}$	.996$_{.000}$
GBDT	.970$_{.048}$	.988$_{.022}$	.988$_{.026}$	.993$_{.014}$	.995$_{.001}$	**.997**$_{.000}$
DeepSets	.992$_{.025}$	**.995**$_{.004}$	.994$_{.004}$	**.995**$_{.000}$	.995$_{.001}$	**.997**$_{.000}$
LOCUS-GNN scratch	.987$_{.021}$	.992$_{.019}$	.992$_{.015}$	**.995**$_{.001}$	.994$_{.001}$	**.997**$_{.001}$
LOCUS probe	.983$_{.020}$	.991$_{.013}$	.993$_{.013}$	.994$_{.001}$	.994$_{.002}$	.996$_{.001}$
LOCUS + CV head	.982$_{.004}$	.990$_{.008}$	.993$_{.008}$	.994$_{.001}$	.995$_{.000}$	.996$_{.000}$
Priors-only LR	**.994**$_{.000}$	.994$_{.000}$	.994$_{.000}$	.994$_{.000}$	.994$_{.000}$	.995$_{.000}$
LOCUS	.989$_{.007}$	.993$_{.005}$	.993$_{.006}$	.994$_{.001}$	.993$_{.001}$	.996$_{.001}$

Table S23. SROIE: auroc as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.859	.859	.859	.859	.859	.859
ConfidNet-style	.815	.867	.912	.941	.948	.974
Flat-LR	.986$_{.010}$	.991$_{.003}$	.990$_{.003}$	.993$_{.003}$	.994$_{.001}$	.995$_{.000}$
CharEv-LR	.991$_{.010}$	.990$_{.011}$	.990$_{.004}$	**.994**$_{.002}$	**.995**$_{.001}$	.996$_{.000}$
GBDT	.978$_{.017}$	.986$_{.028}$	.991$_{.017}$	**.994**$_{.003}$	**.995**$_{.002}$	**.997**$_{.000}$
DeepSets	.983$_{.019}$	.985$_{.013}$	.992$_{.004}$	.993$_{.002}$	.993$_{.001}$	.996$_{.000}$
LOCUS-GNN scratch	.970$_{.030}$	.977$_{.009}$	.990$_{.005}$	.992$_{.002}$	.993$_{.003}$	.995$_{.000}$
LOCUS probe	.991$_{.004}$	**.992**$_{.001}$	**.993**$_{.002}$	.993$_{.001}$	.994$_{.001}$	.996$_{.000}$
LOCUS + CV head	.976$_{.004}$	.986$_{.003}$	.990$_{.004}$	.993$_{.001}$	.993$_{.001}$	.996$_{.000}$
Priors-only LR	**.992**$_{.001}$	**.992**$_{.000}$	**.993**$_{.000}$	.993$_{.000}$	.993$_{.000}$	.993$_{.000}$
LOCUS	.984$_{.011}$	.989$_{.003}$	.992$_{.001}$	.992$_{.001}$	.993$_{.000}$	.995$_{.001}$

Table S24. IAM: auroc as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.878	.878	.878	.878	.878	.878
ConfidNet-style	.835	.849	.861	.876	.894	.917
Flat-LR	.961$_{.010}$	.961$_{.008}$	.970$_{.008}$	.973$_{.006}$	.976$_{.002}$	.982$_{.000}$
CharEv-LR	.927$_{.011}$	.925$_{.011}$	.959$_{.011}$	.970$_{.002}$	.974$_{.003}$	.978$_{.000}$
GBDT	.929$_{.027}$	.948$_{.081}$	.964$_{.012}$	.973$_{.002}$	.976$_{.006}$	**.986**$_{.000}$
DeepSets	**.969**$_{.012}$	.960$_{.021}$	**.972**$_{.014}$	.975$_{.010}$	**.979**$_{.005}$	**.986**$_{.001}$
LOCUS-GNN scratch	.958$_{.016}$	.957$_{.014}$	.969$_{.013}$	.972$_{.009}$	.977$_{.008}$	**.986**$_{.001}$
LOCUS probe	.953$_{.014}$	.957$_{.014}$	.970$_{.007}$	**.976**$_{.001}$	**.979**$_{.001}$	.985$_{.001}$
LOCUS + CV head	.946$_{.020}$	.960$_{.022}$	.969$_{.003}$	.973$_{.004}$	.978$_{.003}$	.985$_{.001}$
Priors-only LR	.950$_{.011}$	**.964**$_{.006}$	.963$_{.005}$	.965$_{.001}$	.965$_{.000}$	.965$_{.000}$
LOCUS	.953$_{.018}$	.963$_{.016}$	.969$_{.007}$	.974$_{.005}$	.978$_{.004}$	**.986**$_{.000}$

Table S25. IIIT5K: auroc as a function of k (labelled words per class). Ensemble of five seeds; subscript: std. of single-seed runs.

Method	k=5	k=10	k=25	k=50	k=100	full
Engine MSP conf.	.938	.938	.938	.938	.938	.938
ConfidNet-style	.982	.981	.980	.984	.985	.988
Flat-LR	.990$_{.002}$	.991$_{.001}$	.991$_{.002}$	.992$_{.002}$	.993$_{.001}$	.994$_{.000}$
CharEv-LR	.990$_{.002}$	.991$_{.000}$	.993$_{.006}$	**.994**$_{.002}$	.994$_{.002}$	**.997**$_{.000}$
GBDT	.976$_{.012}$	.982$_{.012}$	.990$_{.026}$	**.994**$_{.003}$	**.996**$_{.000}$	.995$_{.000}$
DeepSets	.990$_{.003}$	.988$_{.007}$	.990$_{.005}$	**.994**$_{.005}$	.995$_{.003}$	.995$_{.001}$
LOCUS-GNN scratch	.987$_{.004}$	.987$_{.007}$	.990$_{.002}$	**.994**$_{.001}$	.995$_{.001}$	.995$_{.001}$
LOCUS probe	**.991**$_{.002}$	**.992**$_{.001}$	.993$_{.001}$	.993$_{.001}$	.995$_{.002}$	.996$_{.000}$
LOCUS + CV head	.984$_{.002}$	.985$_{.006}$	**.994**$_{.002}$	**.994**$_{.002}$	.995$_{.001}$	.995$_{.001}$
Priors-only LR	**.991**$_{.000}$	.991$_{.000}$	.991$_{.000}$	.991$_{.000}$	.991$_{.000}$	.991$_{.001}$
LOCUS	.986$_{.002}$	.989$_{.002}$	.991$_{.002}$	.992$_{.001}$	.994$_{.002}$	.995$_{.001}$

Table S26. All metrics on FUNSD at $k = 25$ (ensemble of five seeds for learned methods; mean over seeds for threshold baselines). †oracle threshold tuned on test.

Method	Macro-F1	$F1_{err}$	Acc.	AUROC	$AUPRC_{err}$	AURC↓	Cov@95
Engine MSP confidence	.756	.734	.759	.861	.750	.144	.346
ConfidNet-style learned conf. [9]	.771	.723	.781	.845	.805	.164	.053
Two-engine voting	.609	.641	.612	.680	.479	.220	.021
Re-OCR: TrOCR-base [5]	.605	.636	.608	.675	.475	.229	.000
Re-OCR: Florence-2 [6]	.647	.670	.649	.715	.508	.198	.311
Re-OCR: PARSeq [4]	.946	.931	.951	.939	.907	.125	.006
Re-OCR: PARSeq, graded	.946	.931	.951	**.976**	.970	**.090**	**.646**
VLM judge: Qwen2.5-VL-7B [7]	.760	.683	.785	.846	.764	.153	.147
VLM judge, oracle threshold†	.762	.684	.789	.846	.764	.153	.147
LR on flattened graph evidence	**.948**	**.933**	**.952**	.975	**.973**	.092	.645
LR on character evidence	.935	.916	.940	.973	.969	.094	.631
GBDT on all evidence	.922	.903	.927	.969	.964	.098	.625
DeepSets over rivals (no edges)	.942	.926	.947	.971	.968	.094	.639
LOCUS-GNN from scratch	.933	.914	.938	.969	.966	.095	.630
LOCUS zero-shot (no labels)	.918	.890	.927	.949	.953	.110	.270
LOCUS (k-shot)	.943	.926	.948	.974	.971	.092	.628
LOCUS + CV-selected head	.942	.926	.947	.972	.971	.094	.642
LOCUS linear probe	.946	.930	.950	.975	**.973**	.093	.639
Fusion of label-free priors (LR)	.947	.932	**.952**	.966	.968	.101	.643
LOCUS rule (zero-shot, ρ margin)	.866	.836	.873	.943	.939	.110	.471

Table S27. All metrics on CORD at $k = 25$ (ensemble of five seeds for learned methods; mean over seeds for threshold baselines). †oracle threshold tuned on test.

Method	Macro-F1	$F1_{err}$	Acc.	AUROC	$AUPRC_{err}$	AURC↓	Cov@95
Engine MSP confidence	.777	.759	.781	.892	.839	.140	.408
ConfidNet-style learned conf. [9]	.842	.818	.846	.929	.897	.124	.465
Two-engine voting	.772	.775	.772	.807	.635	.170	.409
Re-OCR: TrOCR-base [5]	.811	.807	.812	.841	.679	.154	.447
Re-OCR: Florence-2 [6]	.615	.677	.625	.685	.514	.233	.249
Re-OCR: PARSeq [4]	.973	.968	.974	.975	.945	.109	.615
Re-OCR: PARSeq, graded	.973	.968	.974	.986	.970	.099	.614
VLM judge: Qwen2.5-VL-7B [7]	.891	.862	.899	.945	.943	.127	.351
VLM judge, oracle threshold†	.902	.879	.908	.945	.943	.127	.351
LR on flattened graph evidence	.955	.945	.958	**.995**	**.995**	**.096**	.625
LR on character evidence	.971	.965	.972	.992	.991	.098	.624
GBDT on all evidence	.961	.953	.963	.988	.981	.100	.621
DeepSets over rivals (no edges)	.977	.972	.978	.994	.994	.097	**.627**
LOCUS-GNN from scratch	.969	.963	.970	.992	.991	.098	.624
LOCUS zero-shot (no labels)	.904	.887	.907	.977	.972	.104	.551
LOCUS (k-shot)	**.978**	**.974**	**.979**	.993	.992	.097	.626
LOCUS + CV-selected head	.962	.953	.963	.993	.993	.097	.623
LOCUS linear probe	.961	.952	.963	.993	.992	.099	.625
Fusion of label-free priors (LR)	**.978**	**.974**	**.979**	.994	.994	.097	.626
LOCUS rule (zero-shot, ρ margin)	.934	.923	.935	.991	.990	.098	.618

Table S28. All metrics on SROIE at $k = 25$ (ensemble of five seeds for learned methods; mean over seeds for threshold baselines). †oracle threshold tuned on test.

Method	Macro-F1	$F1_{err}$	Acc.	AUROC	$AUPRC_{err}$	AURC↓	Cov@95
Engine MSP confidence	.780	.766	.781	.859	.826	.211	.091
ConfidNet-style learned conf. [9]	.826	.822	.826	.912	.903	.184	.233
Two-engine voting	.976	.975	.976	.976	.963	.152	.547
Re-OCR: TrOCR-base [5]	**.979**	**.977**	**.979**	.978	.969	.151	.550
Re-OCR: Florence-2 [6]	.478	.676	.553	.578	.512	.377	.092
Re-OCR: PARSeq [4]	.742	.787	.750	.763	.653	.246	.305
Re-OCR: PARSeq, graded	.742	.787	.750	.760	.681	.242	.304
VLM judge: Qwen2.5-VL-7B [7]	.866	.845	.870	.941	.948	.172	.238
VLM judge, oracle threshold†	.883	.870	.885	.941	.948	.172	.238
LR on flattened graph evidence	.955	.951	.956	.990	.991	.139	.546
LR on character evidence	.959	.956	.960	.990	.991	.139	.545
GBDT on all evidence	.962	.959	.962	.991	.991	**.138**	.548
DeepSets over rivals (no edges)	.968	.965	.968	.992	.993	**.138**	.552
LOCUS-GNN from scratch	.961	.958	.962	.990	.992	.139	.550
LOCUS zero-shot (no labels)	.798	.823	.801	.966	.968	.152	.401
LOCUS (k-shot)	.973	.971	.973	.992	.993	.139	.553
LOCUS + CV-selected head	.964	.961	.964	.990	.991	.139	.548
LOCUS linear probe	.973	.971	.973	**.993**	**.994**	**.138**	**.555**
Fusion of label-free priors (LR)	.978	**.977**	**.979**	**.993**	**.994**	**.138**	.553
LOCUS rule (zero-shot, ρ margin)	.910	.910	.911	.981	.981	.144	.507

Table S29. All metrics on IAM at $k = 25$ (ensemble of five seeds for learned methods; mean over seeds for threshold baselines). †oracle threshold tuned on test.

Method	Macro-F1	$F1_{err}$	Acc.	AUROC	$AUPRC_{err}$	AURC↓	Cov@95
Engine MSP confidence	.802	.791	.804	.878	.859	.203	.091
ConfidNet-style learned conf. [9]	.801	.784	.803	.861	.865	.226	.007
Two-engine voting	.759	.769	.760	.767	.663	.272	.001
Re-OCR: TrOCR-base [5]	.758	.769	.759	.766	.661	.273	.001
Re-OCR: Florence-2 [6]	.369	.643	.488	.522	.475	.424	.001
Re-OCR: PARSeq [4]	.879	.881	.879	.885	.799	.186	.455
Re-OCR: PARSeq, graded	.879	.881	.879	.914	.877	.168	.454
VLM judge: Qwen2.5-VL-7B [7]	.700	.713	.700	.788	.765	.250	.052
VLM judge, oracle threshold†	.719	.694	.722	.788	.765	.250	.052
LR on flattened graph evidence	.904	.895	.905	.970	.968	**.146**	.483
LR on character evidence	.887	.878	.888	.959	.956	.152	.446
GBDT on all evidence	.893	.884	.894	.964	.961	.150	.467
DeepSets over rivals (no edges)	**.909**	**.901**	**.910**	**.972**	**.971**	.147	**.497**
LOCUS-GNN from scratch	.908	.899	.909	.969	.970	.149	.482
LOCUS zero-shot (no labels)	.825	.814	.826	.909	.895	.180	.346
LOCUS (k-shot)	.906	.896	.907	.969	.968	.149	.486
LOCUS + CV-selected head	.905	.898	.906	.969	.967	**.146**	.490
LOCUS linear probe	.904	.895	.905	.970	.970	.148	.473
Fusion of label-free priors (LR)	.898	.891	.899	.963	.959	.149	.481
LOCUS rule (zero-shot, ρ margin)	.844	.838	.844	.933	.934	.167	.330

Table S30. All metrics on IIIT5K at $k = 25$ (ensemble of five seeds for learned methods; mean over seeds for threshold baselines). †oracle threshold tuned on test.

Method	Macro-F1	$F1_{err}$	Acc.	AUROC	$AUPRC_{err}$	AURC↓	Cov@95
Engine MSP confidence	.788	.738	.801	.938	.857	.069	.631
ConfidNet-style learned conf. [9]	.922	.892	.933	.980	.958	.054	.726
Two-engine voting	.862	.819	.876	.905	.696	.076	.635
Re-OCR: TrOCR-base [5]	.952	.934	.960	.966	.879	.058	.715
Re-OCR: Florence-2 [6]	.823	.776	.836	.879	.635	.083	.595
Re-OCR: PARSeq [4]	**.993**	**.990**	**.994**	.991	.984	.050	.746
Re-OCR: PARSeq, graded	**.993**	**.990**	**.994**	**.995**	**.994**	**.049**	.747
VLM judge: Qwen2.5-VL-7B [7]	.888	.850	.901	.974	.943	.056	.716
VLM judge, oracle threshold†	.928	.899	.940	.974	.943	.056	.716
LR on flattened graph evidence	.969	.956	.974	.991	.988	.052	.745
LR on character evidence	.965	.951	.971	.993	.990	.050	.745
GBDT on all evidence	.955	.937	.962	.990	.987	.052	.746
DeepSets over rivals (no edges)	.965	.951	.971	.990	.987	.052	.747
LOCUS-GNN from scratch	.962	.946	.968	.990	.987	.052	.747
LOCUS zero-shot (no labels)	.968	.955	.973	.983	.954	.054	.739
LOCUS (k-shot)	.971	.959	.976	.991	.985	.052	.744
LOCUS + CV-selected head	.971	.960	.976	.994	.992	**.049**	.747
LOCUS linear probe	.968	.956	.974	.993	.991	.051	**.748**
Fusion of label-free priors (LR)	.981	.974	.985	.991	.985	.052	.747
LOCUS rule (zero-shot, ρ margin)	.790	.742	.801	.981	.963	.055	.735